

HW 4

Collab:
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Townsend 4.4 (5 pts)

A beam of spin- $\frac{1}{2}$ particles with speed v_0 passes through a series of two SGz devices. The first SGz device transmits particles with $S_z = \hbar/2$ and filters out particles with $S_z = -\hbar/2$. The second SGz device transmits particles with $S_z = -\hbar/2$ and filters out particles with $S_z = \hbar/2$. Between the two devices is a region of length l_0 in which there is a uniform magnetic field B_0 pointing in the x direction. Determine the smallest value of l_0 such that exactly 25 percent of the particles transmitted by the first SGz device are transmitted by the second device. Express your result in terms of $\omega_0 = egB_0/2mc$ and v_0 .

The Hamiltonian in a magnetic field is

$$\hat{H} = \omega_0 \hat{S}_x \quad (1)$$

It travels for a time $t = \frac{l_0}{v_0}$. The field is $B_0 \hat{i}$.

We need to find the time evolution from $\hat{H}$ on $|-z\rangle$. We have the representation of $|-z\rangle$ in the S_x basis as:

$$|-z\rangle = \frac{1}{\sqrt{2}}(|+x\rangle - |-x\rangle) \quad (2)$$

$$\langle +z| = \frac{1}{\sqrt{2}}(\langle +x| + \langle -x|) \quad (3)$$

So we have

$$U(t)|-z\rangle = \frac{1}{\sqrt{2}}e^{-i\omega_0\hat{S}_x\frac{t}{\hbar}}(|+x\rangle - |-x\rangle) \quad (4)$$

$$= \frac{1}{\sqrt{2}}\left(e^{-\frac{i\omega_0}{2}t}|+x\rangle - e^{\frac{i\omega_0}{2}t}|-x\rangle\right) \quad (5)$$

$$(6)$$

We can now find

$$\langle +z|\hat{H}|-z\rangle = \frac{1}{2}\left(e^{-\frac{i\omega_0}{2}t} - e^{\frac{i\omega_0}{2}t}\right) \quad (7)$$

$$= i \sin\left(\frac{\omega_0}{2}t\right) \quad (8)$$

We need

$$|\langle +z | \hat{H} | -z \rangle|^2 = \frac{1}{4} \quad (9)$$

$$\sin^2\left(\frac{\omega_0}{2}t\right) = \frac{1}{4} \quad (10)$$

$$\sin\left(\frac{\omega_0}{2}t\right) = \frac{1}{2} \quad (11)$$

$$\frac{\omega_0}{2}t = \frac{\pi}{6} \quad (12)$$

$$t = \frac{\pi}{3\omega_0} \quad (13)$$

$$t = \frac{l_0}{v_0} \quad (14)$$

We therefore have

$$l_0 = \frac{v_0 \pi}{3\omega_0} \quad (15)$$

Townsend 4.5 (5 pts)

Note that this problem is probably the hardest and certainly the one with the most algebra and half-angle trig identities. It might be best to save this until the end and use Mathematica to simplify things. Mathematica isn't good at taking complex conjugates of functions because it assumes that every variable could be complex and it's hesitant to simplify due to rare branch-cut issues. If I want to take the magnitude squared of something, I just copy it, and paste two copies, and change the second one's i's into minus i's.

A beam of spin- $\frac{1}{2}$ particles in the $|+z\rangle$ state enters a uniform magnetic field B_0 in the x - z plane oriented at an angle θ with respect to the z axis. At time T later, the particles enter an SGy device. What is the probability the particles will be found with $S_y = \hbar/2$? Check your result by evaluating the special cases $\theta = 0$ and $\theta = \pi/2$.

1 Advice from annika

1.1 My advice is "I think i mighta done it wrong" but i have an answer

don't find $\langle +y |$, plug in $+z$ and $-z$ in terms of $-n$. We already found z in terms of the N kets, find $-z$ in a similar way. Everything is real

2 I decided to take the L, the warning scares me <3

Townsend 4.8 (5 pts)

A spin- $\frac{1}{2}$ particle, initially in a state with $S_n = \hbar/2$ with $\mathbf{n} = \sin\theta \mathbf{i} + \cos\theta \mathbf{k}$, is in a constant magnetic field B_0 in the z direction. Determine the state of the particle at time t and determine how $\langle S_x \rangle$, $\langle S_y \rangle$, and $\langle S_z \rangle$ vary with time.

The Hamiltonian for magnetic field $B = B_0 \mathbf{K}$ is

$$\hat{H} = \frac{ge}{2mc} \hat{S}_z B_0 = \omega_0 \hat{S}_z \quad (16)$$

We know the eigenvalues for the magnetic field Hamiltonian

$$\hat{H}|+z\rangle = \omega_0 \hat{S}_z |+z\rangle = \hbar \frac{\omega_0}{2} |+z\rangle \quad (17)$$

$$\hat{H}|-z\rangle = \omega_0 \hat{S}_z |-z\rangle = -\hbar \frac{\omega_0}{2} |-z\rangle \quad (18)$$

We also know that in the z basis,

$$|n\rangle = \cos \frac{\theta}{2} |+z\rangle + e^{i\phi} \sin \frac{\theta}{2} |-z\rangle \quad (19)$$

Here $\phi = 0$.

Therefore, we have

$$|n(t)\rangle = e^{-i\frac{\omega_0}{2}t} \cos \frac{\theta}{2} |+z\rangle + e^{i\frac{\omega_0}{2}t} \sin \frac{\theta}{2} |-z\rangle \quad (20)$$

Now we have to write out the operators in the $|z\rangle$ basis.

2.0.1 $\langle S_z \rangle$

$$S_z = \frac{\hbar}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad (21)$$

$$\langle n(t) | S_z | n(t) \rangle \quad (22)$$

$$= \langle n(t) | \left(\frac{\hbar}{2} e^{-i\omega_0 t} \cos \frac{\theta}{2} |+z\rangle - \frac{\hbar}{2} e^{i\omega_0 t} \sin \frac{\theta}{2} |-z\rangle \right) \quad (23)$$

$$\langle S_z \rangle = \frac{\hbar}{2} \left(\cos^2 \left(\frac{\theta}{2} \right) - \sin^2 \left(\frac{\theta}{2} \right) \right) \quad (24)$$

2.0.2 $\langle S_x \rangle$ and $\langle S_y \rangle$

We have to find this from the raising and lowering operators.

$$S_x = \frac{\hat{S}_+ + \hat{S}_-}{2} \quad (25)$$

$$S_y = \frac{\hat{S}_+ - \hat{S}_-}{2i} \quad (26)$$

We can find the time average of $\hat{S}_+$ and $\hat{S}_-$, then combine them to get the expectation values (that feels valid since we're just adding)

$$\hat{S}_+ = \hbar | +z \rangle \langle -z | \quad (27)$$

$$\hat{S}_- = \hbar | -z \rangle \langle +z | \quad (28)$$

$$(29)$$

$$|n(t)\rangle = e^{-i\frac{\omega_0}{2}t} \cos \frac{\theta}{2} | +z \rangle + e^{i\frac{\omega_0}{2}t} \sin \frac{\theta}{2} | -z \rangle \quad (30)$$

So we have

$$\langle \hat{S}_+ \rangle = \langle n | \hbar e^{i\frac{\omega_0}{2}t} \sin \frac{\theta}{2} | +z \rangle \quad (31)$$

$$= \hbar e^{i\omega_0 t} \cos \frac{\theta}{2} \sin \frac{\theta}{2} \quad (32)$$

We can do the same with $\hat{S}_-$:

$$\langle \hat{S}_- \rangle = \langle n | \hbar e^{-i\frac{\omega_0}{2}t} \cos \frac{\theta}{2} | -z \rangle \quad (33)$$

$$= \hbar e^{-i\omega_0 t} \cos \frac{\theta}{2} \sin \frac{\theta}{2} \quad (34)$$

We can now find the last expectation values

$$S_x = \frac{\hat{S}_+ + \hat{S}_-}{2} \quad (35)$$

$$= \frac{\hbar}{2} \cos \frac{\theta}{2} \sin \frac{\theta}{2} (e^{i\omega_0 t} + e^{-i\omega_0 t}) \quad (36)$$

$$= \hbar \cos \frac{\theta}{2} \sin \frac{\theta}{2} \cos(\omega_0 t) \quad (37)$$

$$(38)$$

And

$$S_y = \frac{\hat{S}_+ - \hat{S}_-}{2i} \quad (39)$$

$$= \hbar \cos \frac{\theta}{2} \sin \frac{\theta}{2} \sin(\omega_0 t) \quad (40)$$

Townsend 4.12 (5 pts)

A particle with intrinsic spin one is placed in a constant external magnetic field B_0 in the x direction. The initial spin state of the particle is $|\psi(0)\rangle = |1, 1\rangle$, that is, a state with $S_z = \hbar$. Take the spin Hamiltonian to be

$$\hat{H} = \omega_0 \hat{S}_x$$

and determine the probability that the particle is in the state $|1, -1\rangle$ at time t . *Suggestion:* If you haven't already done so, you should first work out Problem 3.15 to determine the eigenstates of $\hat{S}_x$ for a spin-1 particle in terms of the eigenstates of $\hat{S}_z$.

We can write the $\hat{S}_x$ operator in the spin 1 $|+z\rangle$ basis

$$\hat{S}_x = \frac{\hbar}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad (41)$$

The eigenvalues are $\hbar, 0, -\hbar$. We can now solve for the vectors, which I label in order as $|1\rangle, |2\rangle, |3\rangle$

$$\frac{1}{2} \begin{bmatrix} 1 \\ \sqrt{2} \\ 1 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \frac{1}{2} \begin{bmatrix} 1 \\ -\sqrt{2} \\ 1 \end{bmatrix} \quad (42)$$

We can find the projection of $|1, 1\rangle$ in the S_x eigenstates, and do the same for the $|1, -1\rangle$ state.

$$\langle 1, 1 |_x |1, 1\rangle_z = \frac{1}{2} [1, \sqrt{2}, 1] \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \frac{1}{2} \quad (43)$$

Similarly, for others we are just keeping the first component and the amplitude. This gets us

$$|1, 1\rangle_z = \frac{1}{2} |1\rangle + \frac{1}{\sqrt{2}} |2\rangle + \frac{1}{2} |3\rangle \quad (44)$$

We can do the same for the final state, $|1, -1\rangle$

$$|1, -1\rangle_z = \frac{1}{2} |1\rangle - \frac{1}{\sqrt{2}} |2\rangle + \frac{1}{2} |3\rangle \quad (45)$$

We can now time evolve, with the individual eigenvalues for each basis ket

$$|\psi(t)\rangle_z = \hat{H} |1, 1\rangle = e^{-i\omega_0 t} \frac{1}{4} \begin{bmatrix} 1 \\ \sqrt{2} \\ -1 \end{bmatrix} + \frac{1}{2} e^0 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + \frac{1}{4} e^{i\omega_0 t} \begin{bmatrix} 1 \\ -\sqrt{2} \\ 1 \end{bmatrix} \quad (46)$$

This collapses down to

$$|\psi(t)\rangle_z = \begin{bmatrix} \frac{e^{-i\omega_0 t}}{4} + \frac{1}{2} + \frac{e^{i\omega_0 t}}{4} \\ \sqrt{2} \frac{e^{-i\omega_0 t}}{4} - \sqrt{2} \frac{e^{i\omega_0 t}}{4} \\ \frac{1}{4}(-e^{-i\omega_0 t} + e^{i\omega_0 t}) - \frac{1}{2} \end{bmatrix} \quad (47)$$

which further simplifies down to

$$|\psi(t)\rangle_z = \begin{bmatrix} \frac{1}{2}(1 + \cos \omega_0 t) \\ \sqrt{\frac{1}{2}}(i \sin(\omega_0 t)) \\ \frac{1}{2}(\cos \omega_0 t - 1) \end{bmatrix} \quad (48)$$

We are now asking the question

$$|\langle 1, -1 | \hat{H} | 1, 1 \rangle|^2 \quad (49)$$

$$= \left| [0, 0, 1] \begin{bmatrix} \frac{1}{2}(1 + \cos \omega_0 t) \\ \sqrt{\frac{1}{2}}(i \sin(\omega_0 t)) \\ \frac{1}{2}(\cos \omega_0 t - 1) \end{bmatrix} \right|^2 \quad (50)$$

$$= \left| \frac{1}{2}(\cos \omega_0 t - 1) \right|^2 \quad (51)$$

$$= \frac{1}{4}(\cos^2(\omega_0 t) - 2 \cos \omega_0 t + 1) \quad (52)$$

Townsend 4.13 (5 pts)

Let

$$\begin{pmatrix} E_0 & 0 & A \\ 0 & E_1 & 0 \\ A & 0 & E_0 \end{pmatrix}$$

be the matrix representation of the Hamiltonian for a three-state system with basis states $|1\rangle$, $|2\rangle$, and $|3\rangle$.

(a) If the state of the system at time $t = 0$ is $|\psi(0)\rangle = |2\rangle$, what is $|\psi(t)\rangle$?

(b) If the state of the system at time $t = 0$ is $|\psi(0)\rangle = |3\rangle$, what is $|\psi(t)\rangle$?

First, we need to find the Eigenbasis for this system. Excluding the center row and column, the matrix looks symmetric as

$$\begin{bmatrix} E_0 & A \\ A & E_0 \end{bmatrix} \quad (53)$$

This has the familiar eigenvectors

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ and } \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (54)$$

The third eigenvector has to be orthogonal as the set spans $\mathbb{R}^3$, so we have

$$\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \quad (55)$$

$$H = \begin{bmatrix} E_0 & 0 & A \\ 0 & E_1 & 0 \\ A & 0 & E_0 \end{bmatrix} \quad (56)$$

Applying each, we get the eigenvalues

$$(E_0 + A) \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, E_1 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, (E_0 - A) \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \quad (57)$$

2.1 A

$|2\rangle$ is an eigenstate because we have $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ already.

$$\psi(t) = e^{\frac{-i}{\hbar} E_1 t} |2\rangle \quad (58)$$

2.2 B

$$|3\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad (59)$$

This is the sum of the first and third eigenstates, so we have

$$\psi(t) = \frac{1}{2} \left(e^{\frac{-i}{\hbar} (E_0 + A)t} (|1\rangle - |3\rangle) - e^{\frac{-i}{\hbar} (E_0 - A)t} (|1\rangle + |3\rangle) \right) \quad (60)$$

$$= \frac{1}{2} e^{\frac{-i}{\hbar} E_0 t} \left(e^{\frac{-i}{\hbar} A t} (|1\rangle - |3\rangle) - e^{\frac{i}{\hbar} A t} (|1\rangle + |3\rangle) \right) \quad (61)$$

$$= \frac{1}{2} e^{\frac{-i}{\hbar} E_0 t} \left(i \sin \frac{A}{\hbar} t |1\rangle + \cos \frac{A}{\hbar} t |3\rangle \right) \quad (62)$$

3 old work

We could use the equation

$$(E_1 - \lambda)(E_0 - \lambda)^2 - A^2(E_1 - \lambda) = 0 \quad (63)$$

$$(64)$$

This has the solutions $E_1 = \lambda$, and some quadratic thing.

$$(E_0 - \lambda)^2 = A^2 \quad (65)$$

$$E_0^2 - 2E_0\lambda + \lambda^2 - A^2 = 0 \quad (66)$$

$$\lambda^2 - 2E_0\lambda + (E_0^2 - A^2) = 0 \quad (67)$$

$$\lambda = \sqrt{\quad} \quad (68)$$

4 OLD WORK 2

given $\psi(0) = |2\rangle$, this becomes

$$\begin{pmatrix} E_0 & 0 & A \\ 0 & E_1 & 0 \\ A & 0 & E_0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad (69)$$

$$= \begin{pmatrix} 0 \\ E_1 \\ 0 \end{pmatrix} \quad (70)$$

This means that evolving in time is

$$|\psi(t)\rangle = e^{\frac{-i}{\hbar}E_1 t} |2\rangle \quad (71)$$

4.1 B

We get

$$H|\psi\rangle = \begin{pmatrix} A \\ 0 \\ E_0 \end{pmatrix} \quad (72)$$

Therefore,

